Practical 7: Dimensionality Reduction## Practical Information| Attribute | Details ||----------|---------|| **Practical Number** | 7 || **Title** | Dimensionality Reduction || **Duration** | 2 Hours || **Course Outcomes (COs)** | CO3, CO4 || **Description** | Apply dimensionality reduction techniques to high-dimensional datasets. |## Key Topics Covered- PCA- LDA- t-SNE- UMAP- Feature selection- Autoencoder---## Instructions for Students1. **Complete all tasks** outlined in this practical2. **Write your code** in the cells provided below3. **Add comments** to explain your code4. **Include proper output** (prints, visualizations, etc.)5. **Submit your completed notebook** before the deadline---## Your Work Starts HereUse the cell(s) below to write your implementation:

```
# Practical 7: Dimensionality Reduction
# Phase 1: Setup & Dataset

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
from sklearn.manifold import TSNE
from sklearn.feature_selection import SelectKBest, f_classif
```

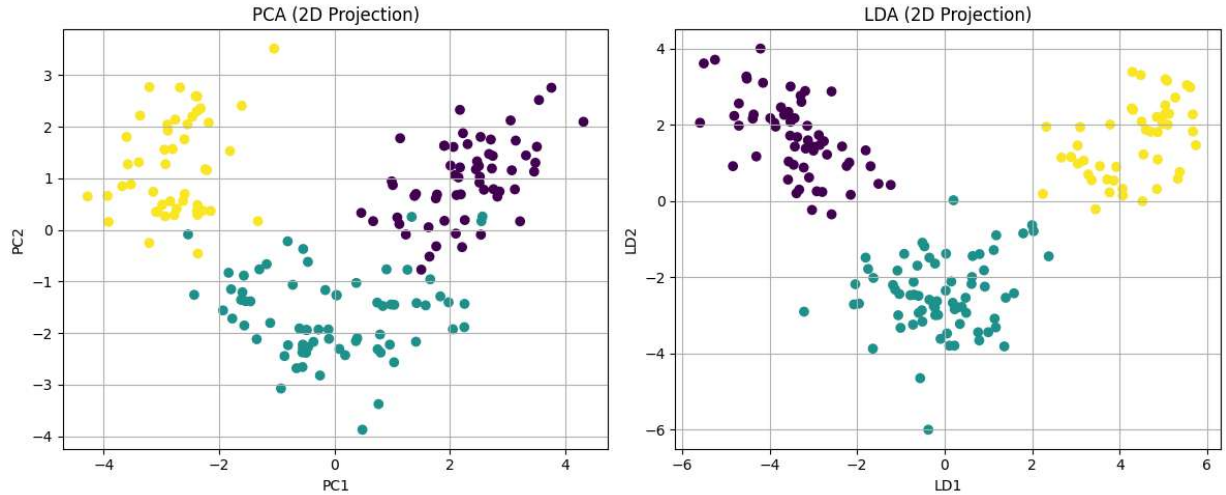
```
# Load dataset
data = load_wine()
X = data.data
y = data.target
feature_names = data.feature_names

print("Dataset shape:", X.shape)
print("Number of classes:", len(np.unique(y)))

# Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("✅ Dataset loaded and scaled successfully!")
```

```
Dataset shape: (178, 13)
Number of classes: 3
✅ Dataset loaded and scaled successfully!
```

```
# Phase 2: PCA & LDA

# PCA (2D)
pca = PCA(n_components=2, random_state=42)
X_pca = pca.fit_transform(X_scaled)

print("PCA Explained Variance Ratio:", pca.explained_variance_ratio_)

# LDA (2D)
lda = LDA(n_components=2)
X_lda = lda.fit_transform(X_scaled, y)

# Plot PCA and LDA
plt.figure(figsize=(12, 5))

# PCA plot
plt.subplot(1, 2, 1)
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.title("PCA (2D Projection)")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.grid(True)

# LDA plot
plt.subplot(1, 2, 2)
plt.scatter(X_lda[:, 0], X_lda[:, 1], c=y)
plt.title("LDA (2D Projection)")
plt.xlabel("LD1")
plt.ylabel("LD2")
plt.grid(True)

plt.tight_layout()
```

```
plt.show()

print("✅ PCA and LDA visualization completed!")
```

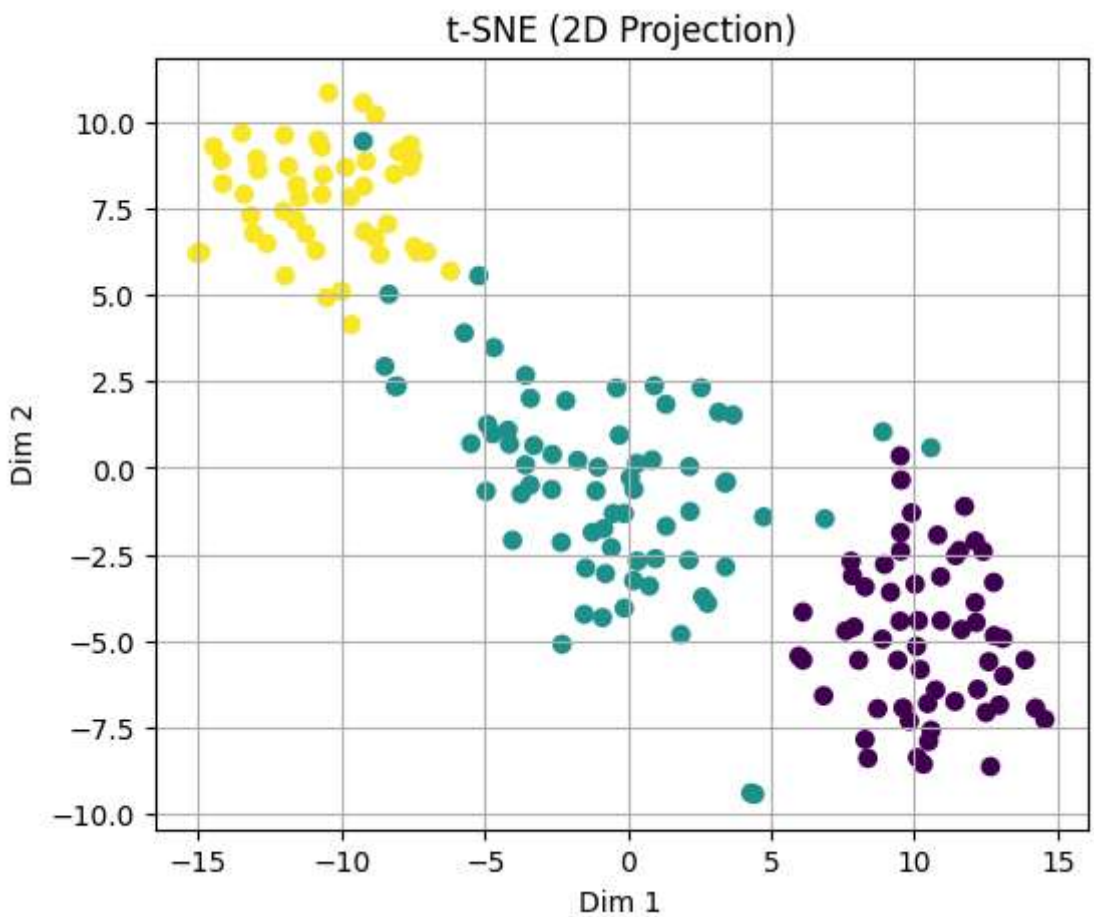
```
PCA Explained Variance Ratio: [0.36198848 0.1920749 ]
```




```
✅ PCA and LDA visualization completed!
```

```
# Phase 3: t-SNE + Feature Selection

# t-SNE (2D)
tsne = TSNE(n_components=2, perplexity=30, random_state=42)
X_tsne = tsne.fit_transform(X_scaled)

# Plot t-SNE
plt.figure(figsize=(6, 5))
plt.scatter(X_tsne[:, 0], X_tsne[:, 1], c=y)
plt.title("t-SNE (2D Projection)")
plt.xlabel("Dim 1")
plt.ylabel("Dim 2")
plt.grid(True)
plt.show()

# Feature Selection (Top 5 features)
selector = SelectKBest(score_func=f_classif, k=5)
X_selected = selector.fit_transform(X_scaled, y)
```

```
selected_features = np.array(feature_names)[selector.get_support()]
print("Top 5 Selected Features:")
for f in selected_features:
    print("-", f)

print("\nReduced Feature Shape:", X_selected.shape)
print("✅ t-SNE and Feature Selection completed!")
```




```
Top 5 Selected Features:
- alcohol
- flavanoids
- color_intensity
- od280/od315_of_diluted_wines
- proline

Reduced Feature Shape: (178, 5)
✅ t-SNE and Feature Selection completed!
```